

Fold-to-Fit: Propeller Interaction and Control of a Coaxial-Bicopter Quadrotor

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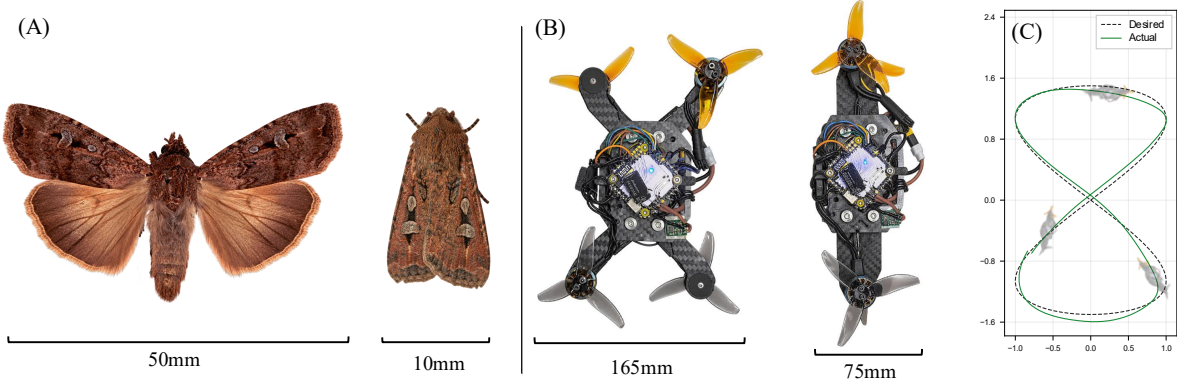


Fig. 1. **Fold to fit.** (A) A bogong moth spans ~ 50 mm in flight and folds its wings to a ~ 10 mm profile. (B) TinyMorph does the same in flight: morphing from the X configuration ($\alpha = 45^\circ$, left) to the fully folded coaxial bicopter ($\alpha = 0^\circ$, right) collapses the rotors into two coaxial pairs, reducing the lateral footprint from 165 mm to 75 mm ($2.2\times$); (C) Closed-loop Figure-8 tracking in the folded configuration using our propeller interaction model.

Abstract—A traditional fixed quadrotor cannot traverse a gap narrower than its own lateral footprint. Gap traversal of a narrower gap might require the quadrotor to reconfigure its airframe on the fly, until its lateral footprint is only limited by its propeller diameter. In this work, we introduce TinyMorph, a 250 g quadrotor that folds in flight into a longitudinal coaxial bicopter configuration, reducing its lateral footprint by $2.2\times$ matching its propeller diameter. Conventional roll control of a quadrotor uses the lateral thrust moment arm, which vanishes in the bicopter configuration. A fixed inward motor tilt, or dihedral, can instead project propeller reaction moment onto the roll axis [1], preserving some roll control. Folding also places each lower rotor in the airflow of an upper rotor, reducing its thrust and torque effectiveness and changing the motor commands required to produce a desired control wrench. We develop an online propeller interaction model that estimates these effects, improving the performance of the quadrotor. A closed form allocation correction ensures that the new motor request respects the desired actuation forces. The method is implemented and evaluated in bicopter flight. In the Figure-8 flights, the propeller interaction enabled run exhibits a 40.9% lower root mean square (RMS) error.

I. INTRODUCTION

The narrowest passage a flying robot can traverse is set by its own cross section. Insects and birds treat these cross sections as a variable. For instance, a bogong moth spanning ~ 50 mm in flight can fold its wings flat against its body to about $0.2\times$ its size (~ 10 mm) and slip into the rock crevices where it aestivates [2], and earwigs fold their rearwings to a tenth or less of the flight area to squeeze through soil and bark [3]. Autonomous gap detection and traversal have also been demonstrated for quadrotors using onboard vision [4]. Such methods determine where and how the vehicle should pass through an opening, but the minimum traversable width

remains bounded by the physical footprint of the aircraft i.e. the geometry itself must change. TinyMorph folds in flight from the X quadrotor configuration into a longitudinal coaxial I-shape bicopter at variable amounts of folding, reducing its lateral width from 165 mm to a minimum of 75 mm i.e. a $2.2\times$ reduction.

There are morphing drones that squeeze through the gaps: folding arms [5], sprung passive hinges [6], and elastic frames [7], [8], but the coaxial bicopter configuration, is where the actuation model changes most. The lateral moment arm that generates roll shrinks as $L \sin \alpha$ and vanishes as the drone morphs to this configuration. Conventional motor mounting would eliminate roll authority in this state; a fixed motor dihedral can instead project propeller reaction moment onto the roll axis [1], preserving some roll control. The lower rotor of each coaxial pair is in the airflow wake of its upper rotor and no longer matches the thrust and reaction-moment effectiveness assumed by an isolated rotor model [9], [10]. These effects dominate in micro drones not only due to propeller interaction with each other but also due to lesser reaction moment provided by small propellers and propeller airframe interaction. A micro vehicle that must retain control at minimal cross section therefore requires both an appropriate folded actuation model and compensation for coaxial interaction.

This paper contributes: (i) an online RPM dependent propeller interaction model that estimates the reduced thrust and torque effectiveness of the overlapped rotors, together with a bounded interaction aware allocation that corrects the baseline motor commands and (ii) an experimental evaluation of the fully folded flight showing that the proposed model improves trajectory tracking, including a 40.9% reduction in three-dimensional RMS error in the figure eight experiment.

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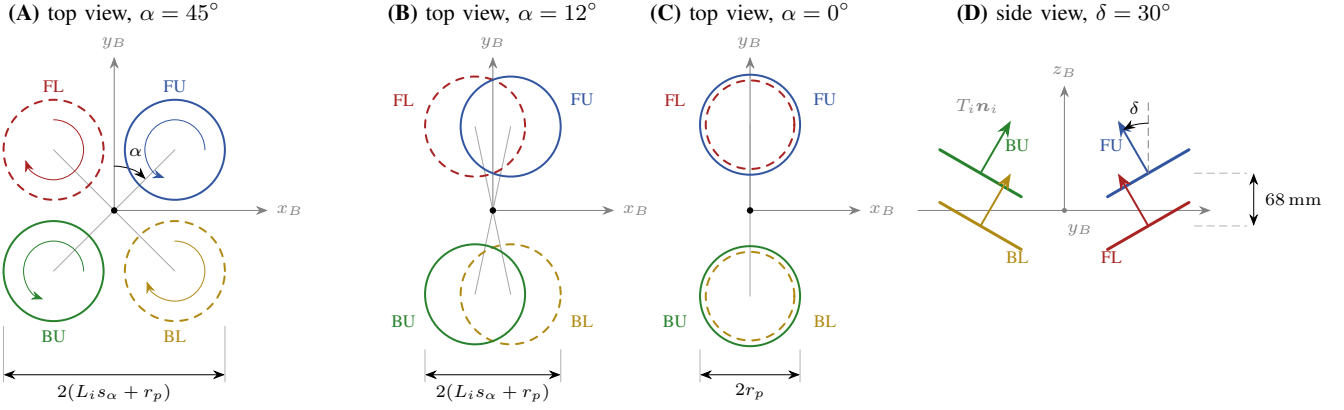


Fig. 2. **Morphing geometry and dihedral tilt.** (A) X configuration, with rotor colors denoting FU, FL, BU, and BL L_i is the length of an arm, r_p is radius of propeller; upper rotors are solid and lower rotors dashed. (B) During morphing, the rotor pairs converge and the lateral moment arm shrinks with $L_i \sin \alpha$. (C) At $\alpha = 0^\circ$, the front and rear rotor pairs overlap along y_B . (D) The motors use a fixed dihedral $\delta = 30^\circ$. Relative to the center of gravity, the upper and lower rotor planes are located at $z_U = +48$ mm and $z_L = -20$ mm, giving a vertical separation of 68 mm.

Although demonstrated on a micro drone platform, the proposed method requires only an electronic speed controller (ESC) motor speed telemetry and a closed form allocation update. It compensates for coaxial interaction without closed loop RPM control or a high fidelity aerodynamic model. This is particularly important in the bicopter configuration, where coaxial losses overlap with reduced control authority.

II. COAXIAL BICOPTER MORPHING MODEL

This section describes how morphing affects the forces and torques available for attitude control. First, we define the rotor geometry and show how morphing eliminates the lateral moment arm used for roll control by conventional quadrotors. Then, we show how a fixed dihedral preserves roll authority by projecting rotor reaction moment onto the roll axis.

A. Morphing Geometry

TinyMorph is a quadrotor whose four arms are mechanically coupled and actuated by a single servo. The linkage moves the front and rear arm pairs symmetrically through the morphing angle α . At $\alpha = 45^\circ$, the rotor hubs form an X configuration; as α decreases, the arms fold toward the longitudinal centerline, reaching front and rear coaxial pairs at $\alpha = 0^\circ$, as illustrated in Fig. 2A–C. The propeller pitch and motor mounting orientation remain fixed. As used here, the motor dihedral δ is the fixed angle between each rotor axis and the vertical body axis z_B . The rotor axes are tilted inward in the vertical plane containing each arm, as shown in Fig. 2D.

The body frame is right-handed, with x_B to the right, y_B forward, and z_B upward; consequently, roll is rotation about y_B . The motors are ordered as (FU, FL, BU, BL), denoting the front-upper, front-lower, back-upper, and back-lower rotors. Let $\mathbf{r}_i(\alpha)$ denote the position of the rotors relative to the centre of gravity (CoG). The sign vectors ℓ and $\mathbf{f}$

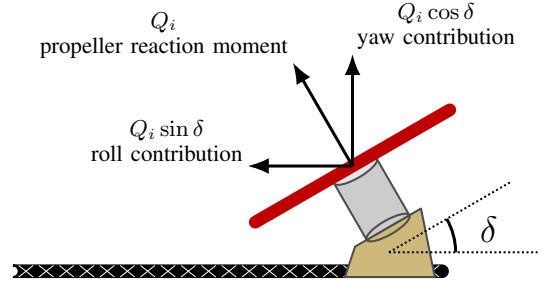


Fig. 3. **Motor dihedral and reaction-moment decomposition.** The propeller reaction moment Q_i , aligned with the inclined rotor axis, has a vertical component $Q_i \cos \delta$ and a component $Q_i \sin \delta$ along its arm.

specify the lateral side and the front back position of each rotor, respectively. The rotor positions are:

$$\begin{aligned} \mathbf{r}_i(\alpha) &= [\ell_i L_i \sin \alpha \quad \mathbf{f}_i L_i \cos \alpha \quad z_i]^T, \\ \ell &= (1, -1, -1, 1), \quad \mathbf{f} = (1, 1, -1, -1), \\ \mathbf{L} &= (6.5, 7.5, 6.5, 7.5) \times 10^{-2} \text{ m}, \\ \mathbf{z} &= (4.8, -2.0, 4.8, -2.0) \times 10^{-2} \text{ m}. \end{aligned} \quad (1)$$

Viewed from above, the FU and BU rotors rotate counterclockwise, while the FL and BL rotors rotate clockwise. Eq. (1) shows that the lateral thrust moment arm of rotor i is proportional to $L_i \sin \alpha$ and therefore vanishes in the fully folded configuration as $\alpha \rightarrow 0$. Thus, conventional differential thrust roll authority decreases as the vehicle folds and is lost at $\alpha = 0^\circ$. This motivates the dihedral based roll mechanism discussed in the following subsection.

B. Recovering Roll Authority with Dihedral Angles

Traditional quadrotors leverage the rotor reaction torque for yaw control. Tilting the rotor axes redirects a component of this reaction moment into the body horizontal plane, where it is distributed between the roll and pitch axes according to α as illustrated in Fig. 3; in the bicopter configuration, it acts entirely along the roll axis. TinyMorph uses a fixed motor dihedral $\delta = 30^\circ$.

Let T_i and Q_i denote the thrust and torque magnitudes of rotor i , respectively. Neglecting the vertical rotor offsets temporarily, the unsigned roll $\bar{\tau}_{R,i}$, unsigned pitch $\bar{\tau}_{P,i}$, and unsigned yaw moment $\bar{\tau}_{Y,i}$ are given by:

$$\begin{bmatrix} \bar{\tau}_{R,i} \\ \bar{\tau}_{P,i} \\ \bar{\tau}_{Y,i} \end{bmatrix} = \begin{bmatrix} L_i T_i \cos \delta \sin \alpha + Q_i \sin \delta \cos \alpha \\ L_i T_i \cos \delta \cos \alpha + Q_i \sin \delta \sin \alpha \\ Q_i \cos \delta \end{bmatrix}. \quad (2)$$

The T_i terms represent rotor thrust acting through moment arms that vary with the morphing angle α . The Q_i terms arise from the projection of the propeller reaction moment onto the body axes. In the fully folded configuration, $\alpha = 0^\circ$, Eq. (2) reduces to

$$\bar{\tau}_{R,i} = Q_i \sin \delta, \quad \bar{\tau}_{P,i} = L_i T_i \cos \delta, \quad \bar{\tau}_{Y,i} = Q_i \cos \delta \quad (3)$$

The vertical rotor offset contributions and rotor dependent signs are included in the complete folded wrench map in Section III.

III. INTERACTION AWARE GEOMETRIC CONTROLLER

Let $R, R_d \in SO(3)$ be the actual and desired body-to-world rotations. Let ω be the measured body angular velocity and ω_d the desired angular velocity expressed in the desired body frame. Attitude is controlled using a geometric controller on $SO(3)$ [11]. The attitude rate error e_R and the angular rate error e_ω are given by

$$\begin{aligned} e_R &= \frac{1}{2}(R^T R_d - R_d^T R)^\vee, \\ e_\omega &= R^T R_d \omega_d - \omega. \end{aligned} \quad (4)$$

where $(\cdot)^\vee$ converts a skew-symmetric matrix into its corresponding vector. The desired torque τ_d is

$$\tau_d = K_R e_R + K_\omega e_\omega + \omega \times J \omega + \hat{F}_z b \quad (5)$$

where K_R, K_ω : attitude and angular rate gains; J : body inertia matrix, $f_{c,i}$: filter cutoff frequency for axis i , $\omega \times J \omega$: rigid body rotational coupling compensation; $\hat{F}_z$: estimated positive thrust component along body z , $\hat{F}_z b$: thrust dependent compensation for persistent moments; $b = [b_R, b_P, b_Y]^T$ contains bias arms.

$$\begin{aligned} K_R &= 10^{-2} \text{diag}(4.0, 1.4, 0.3), \\ K_\omega &= 10^{-3} \text{diag}(6.0, 2.8, 2.0), \\ J &= 10^{-4} \text{diag}(4.68, 1.41, 3.80) \text{kg m}^2, \\ b &= [0, 0, -1.24 \times 10^{-3}]^T \text{m}. \end{aligned} \quad (6)$$

A. RPM dependent coaxial interaction model

For each coaxial pair $p \in \{F, B\}$, define a relative rotor speed state χ_p , where,

$$\chi_p = \frac{\Omega_{U,p}^2}{\Omega_{U,p}^2 + \Omega_{L,p}^2} \quad (7)$$

where Ω_U and Ω_L denotes rotor speed of upper and lower motor respectively. The model reduces the lower rotor's thrust and torque effectiveness independently represented by $\eta_{T,p}$ and $\eta_{Q,p}$ respectively:

$$\eta_{T,p} = 1 - \sigma C_T \chi_p, \quad \eta_{Q,p} = 1 - \sigma C_Q \chi_p. \quad (8)$$

where, σ denotes prescribed rotor overlap factor, $\sigma = 1$ for bi-copter configuration; C_T, C_Q are the configured thrust loss and reaction torque loss coefficients.

$$C_T = 0.40, \quad C_Q = 0.12. \quad (9)$$

B. Geometry dependent force and torque model

Let $\mathbf{u} = [u_{FU}, u_{FL}, u_{BU}, u_{BL}]^T$ denote the normalized, thrust linearized motor requests produced by the baseline mixer. Throughout this section, vector entries follow the rotor ordering $(1, 2, 3, 4) = (FU, FL, BU, BL)$. Each u_i is dimensionless and represents the requested actuation of rotor i ; we call this motor effort. The implemented model is linear in that effort, with RPM dependent coefficients:

$$T_i = t_i u_i, \quad Q_i = q_i u_i. \quad (10)$$

where, t_i and q_i are the thrust and torque effectiveness coefficients of rotor i . The constants t_0 and q_0 denote the nominal effectiveness of a rotor which depend on the motor and propeller used.

For the nominal model, all four rotors are assigned the same effectiveness:

$$\mathbf{t}_{noint} = t_0 [1, 1, 1, 1]^T, \quad \mathbf{q}_{noint} = q_0 [1, 1, 1, 1]^T. \quad (11)$$

For the interaction model, the lower rotor effectiveness is modified:

$$\begin{aligned} \mathbf{t}_{int} &= t_0 [1, \eta_{T,F}, 1, \eta_{T,B}]^T, \\ \mathbf{q}_{int} &= q_0 [1, \eta_{Q,F}, 1, \eta_{Q,B}]^T. \end{aligned} \quad (12)$$

With $s = \sin \delta$ and $c = \cos \delta$, the effective pitch moment arms of the upper and lower rotors are

$$a_U = 0.065c + 0.048s, \quad a_L = 0.075c - 0.020s. \quad (13)$$

At $\delta = 30^\circ$, $a_U = 0.080$ m and $a_L = 0.055$ m.

The force-torque vector also called the control wrench is

$$\mathbf{v} = [F_z \quad \tau_R \quad \tau_P \quad \tau_Y]^T = \mathcal{B}(\mathbf{t}, \mathbf{q}) \mathbf{u}, \quad (14)$$

and $\mathcal{B}(\mathbf{t}, \mathbf{q})$ is the control effectiveness matrix mapping motor commands $\mathbf{u}$ to the control wrench. Where,

$$\mathcal{B}(\mathbf{t}, \mathbf{q}) = \begin{bmatrix} ct(1) & ct(2) & ct(3) & ct(4) \\ sq(1) & -sq(2) & -sq(3) & sq(4) \\ a_U t(1) & a_L t(2) & -a_U t(3) & -a_L t(4) \\ cq(1) & -cq(2) & cq(3) & -cq(4) \end{bmatrix} \quad (15)$$

The nominal and interaction aware wrench matrices are denoted by $\mathbf{B}_0$ and $\mathbf{B}_{int}$ respectively. Where,

$$\mathbf{B}_0 = \mathcal{B}(\mathbf{t}_{noint}, \mathbf{q}_{noint}), \quad \mathbf{B}_{int} = \mathcal{B}(\mathbf{t}_{int}, \mathbf{q}_{int}). \quad (16)$$

C. Interaction compensation while preserving the control wrench

Let $\mathbf{u}_0$ be the feasible motor request produced by the baseline controller and allocator. Its control wrench is $\mathbf{B}_0 \mathbf{u}_0$. Let $\mathbf{u}^*$ be the motor request produced by the propeller interaction controller and allocator. Its control wrench is $\mathbf{B}_{int} \mathbf{u}^*$.

We need to find the motor commands for the propeller interaction controller that produce the same wrench. Therefore,

$$\mathbf{B}_{int} \mathbf{u}^* = \mathbf{B}_0 \mathbf{u}_0 \implies \mathbf{u}^* = \mathbf{B}_{int}^{-1} \mathbf{B}_0 \mathbf{u}_0. \quad (17)$$

IV. RESULTS

A. Experimental Setup

TinyMorph is a custom built morphing quadrotor capable of flight at the I-shape bi-copter configuration. It is built using a MicoAirH743V2 AIO 45A flight controller and the motors and propellers used are T-Motor F1404 4600KV and GEMFAN 3020-3 respectively. The morphing is through a

NSDRC RS-100 servo. The custom controller is built within the ArduPilot custom control framework using ArduCopter V4.7.0. The geometric attitude controller and the interaction aware allocation execute at 400 Hz loop rate while position control is the default ArduPilot position control.

Vehicle pose was measured using a Vicon motion capture system and transmitted to the flight controller via MAVLink at 30 Hz. The Vicon position, velocity, and yaw measurements were transmitted to ArduPilot as external navigation inputs and fused with the onboard inertial measurements. Trajectory setpoints were transmitted to ArduPilot at 20 Hz. Vicon position measurements were also used independently to calculate the trajectory tracking errors reported in Section IV-B.

B. Trajectory Tracking

We evaluated the interaction model in fully folded coaxial bi-copter configuration using Figure-8 and Straight-Line trajectories measured by Vicon. Each comparison contains results with the interaction model disabled and enabled. The controller, gains, aircraft configuration, and trajectory controller were unchanged within each comparison. Since the coaxial interaction model affects both the thrust and torque commanded, three dimensional positional error alone cannot clearly distinguish between the effect of improved collective force produced and the improved attitude control. The trajectories are decomposed into two and three dimensions for this analysis

The Straight-Line experiment is setup with the constant yaw in the direction of motion, which makes it makes the motion predominantly pitch dominated. The Straight-Line experiment repeatedly commanded an approximately 2.9 m diagonal path using 5 s legs. This comparison reduced three-dimensional RMS tracking error from 0.134 m to 0.109 m, an improvement of 18.4%. The lateral deviation seen in Fig. 4 highlights the importance of stabilizing the weak roll channel despite the motion being pitch dominated.

The Figure-8 provides a more challenging test of the same controller. The vehicle must generate alternating lateral accelerations through repeated left and right roll motion, which shows the improvement of the attitude controller with the interaction model enabled. The Figure-8 flights produced lower errors with the interaction model enabled, including an observed 40.9% reduction in three-dimensional RMS error.

The RMS results from Table I show that the majority of the improvement in tracking is due to the interaction aware attitude controller. The larger improvement in Figure-8 is consistent with its greater excitation of the roll channel. The commanded motion repeatedly depends on the rotor torque, whose effectiveness is explicitly corrected by the proposed model.

V. CONCLUSION

We presented an RPM-dependent propeller interaction model and closed-form motor correction tested on TinyMorph in its fully folded configuration. The propeller interaction model reduced the position tracking error in Figure-8 trajectory by 40.9% compared to the no interaction model.

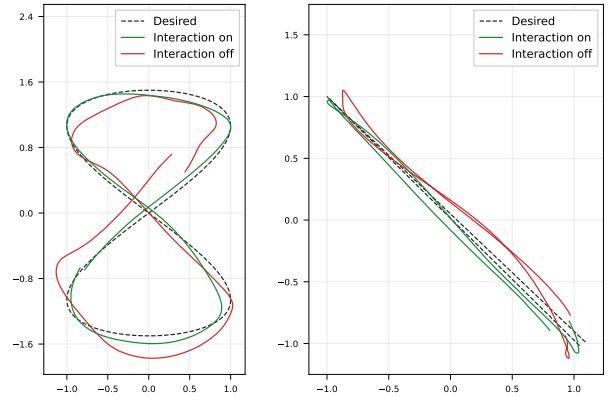


Fig. 4. **Planar trajectory tracking.** Left: Figure-8. Right: Straight-Line. Dashed black denotes the desired trajectory, green denotes interaction model enabled, and red denotes interaction model disabled.

TABLE I
TRAJECTORY TRACKING ERRORS COMPARISON BETWEEN INTERACTION AND BASELINE MODEL.

Trajectory	Method	RMS_z	RMS_{2D}	RMS_{3D}	ΔRMS_{3D}
Figure-8	Baseline	0.0106	0.3151	0.3153	–
Figure-8	Interaction	0.0083	0.1863	0.1864	40.9%
Straight-Line	Baseline	0.0060	0.1338	0.1339	–
Straight-Line	Interaction	0.0055	0.1092	0.1093	18.4%

RMS_z denotes vertical error, RMS_{2D} horizontal planar error, and RMS_{3D} total spatial error. ΔRMS_{3D} is the percentage reduction relative to the corresponding baseline. All values are in meters.

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